

STRIPBOARD TOOLS FOR ELECTRONIC PROTOTYPING

Let the designers revise the component layout as they work towards a neat compact result!

■ ABHIJIT PAUL CHOUDHURY

Stripboard, *aka* Veroboard, is a widely used electronics prototyping board. In PCB designing, the user first places the components, then draws the copper tracks to connect component pins. This is a straightforward process. But, in the case of stripboard, the user has to design the layout by judicious component placement onto the existing copper tracks, cutting the tracks where needed and finally joining distant strips with insulated wires or small two-leaded resistors, diodes, capacitors, etc.

Rick Murray, developer of VeroDes, says, "Traditional PCB design software works on a blank canvas approach where you layout the circuit as you wish. Stripboard, on the other hand, contains the constraints that there are rows of conductive tracks already present on the board. It would seem, to me, that a lot of PCB design software do not cope well with this type of board."

Why is a stripboard software important? The initial placement is critical and many components are variable in length, so the layout becomes a puzzle that can easily go wrong. That's where computer CAD can help.

Roger Lascelles, developer, VeeCAD, says, "Even the simplest stripboard CAD program lets the users revise the layout as they work towards a neat compact result. Another reason for using the stripboard software is that the copper tracks in stripboard are fixed and hence it can sometimes be a real pain to convert schematics into Veroboard."

Before purchasing any stripboard software, "the user should look into the demo version of the software and

try it out. Second aspect to look for in a software is whether the software is being updated from time to time by the company," says Murray. Also, there are many software that are quite expensive, costing around 30 euros. If the user is designing a simple board, there is no need to buy such an expensive software. Rather, there are certain free software that the user can work with.

Given below are some commonly used stripboard software.



VeeCAD is equivalent of a full netlist PCB layout editor like Protel and Orcad. The latest version available is 2.16.1.0.

What's Special?

VeeCAD comes with various features that make designing easy. It lets the user draw schematics and export the netlist. It also allows netlist import from software like Protel, Tango, Orcad PCB 2 and Easy-PC Generic. Moving, rotating and resizing the components become very easy in VeeCAD. Further, the user can label the components. The software comes with auto-router—all the user needs to do is place fixed parts on the board; the auto router places breaks, links and 2-wire parts; and at the end, the user needs to place unrouted items and tidy up. The user can check the design for errors and trace nets. He can also take a printout after the design is complete.

Roger says, "A few of the future features that I am working on are non-numeric pin names, editor copy-paste of components, links and breaks, router improvement and line connection lines

to the pins of a dragged component." He adds, "VeeCAD currently has no bugs as any important bug report filed gets fixed within a day or two."

Users' feedback

VeeCAD is equivalent of a full netlisted PCB layout editor like Protel and Orcad. It reads the user netlist, generates components and helps the user get the layout correct.

VeeCAD has a better symbol library in both free and commercial versions. The commercial version now has rotation of a selected block and improved auto router.

Suggestion

Include blocking of copy-paste of any section of a layout. Also, integrating the rotation of a selected block in demo version would be great. The commercial version has a better auto router, which could be added to the demo version too.

Distributor

So far, there are no distributors. VeeCAD is sold online at a reasonable cost of \$26.26 (for one licence). Both free and commercial versions are supported by email and forum. For more information, visit <http://veecad.com/buy.html>

Home Page: <http://veecad.com/index.html>
Download URL: <http://veecad.com/downloads.html>

Platform Support: Windows



Loch Master software by ABACOM is a developers' tool for stripboard projects. The latest version of the software avail-